



FIELD PROJECT REPORT



DULHASTI POWER STATION KISHTWAR (J&K)



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MECHANICAL ENGINEERING



INDIAN INSTITUTE OF TECHNOLOGY JAMMU



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This internship has been an enriching experience that has enhanced my technical knowl- edge and practical understanding.

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1) INTRODUCTION TO NHPC

NHPC Limited (formerly National Hydroelectric Power Corporation) is a Government of India enterprise established in 1975 with the objective of planning, promoting, and developing hydroelectric power projects across the country. Over the years, NHPC has emerged as one of India's leading organizations in the hydropower sector, contributing significantly to the nation's energy requirements. In addition to hydropower development, NHPC has expanded its activities into renewable energy sectors such as solar, wind, tidal, and geothermal energy.

NHPC is committed to the generation of clean, reliable, and sustainable energy while ensuring environmental conservation and socio-economic development. The organization has successfully executed numerous hydroelectric projects across India and continues to play a vital role in strengthening the nation's power infrastructure. Through the adoption of advanced technologies and efficient project management practices, NHPC has established itself as a premier organization in the field of renewable energy development.

2) DULHASTI POWER STATION

Dulhasti Power Station is a run-of-the-river hydroelectric project with pondage located on the Chenab River in the Kishtwar district of Jammu and Kashmir. The project has an installed capacity of 390 MW, consisting of three generating units of 130 MW each. It is designed to harness the hydropower potential of the Chenab River and contribute significantly to the regional and national power grid.

The project consists of a 65 m high and 186 m long concrete gravity dam, three penstocks, and an underground powerhouse housing the generating units. The power station operates under a net rated head of approximately 207.5 m and is designed to generate about 1907 million units (MU) of electricity annually under dependable operating conditions. The project plays a vital role in meeting the growing demand for clean and renewable energy in India.

Dulhasti Power Station is one of the major hydroelectric projects of NHPC in the northern region and contributes significantly to power generation in the country. The project not only supplies electricity to various states but also supports regional development through infrastructure growth, employment generation, and the promotion of sustainable energy resources.

3) SPECIFICATION

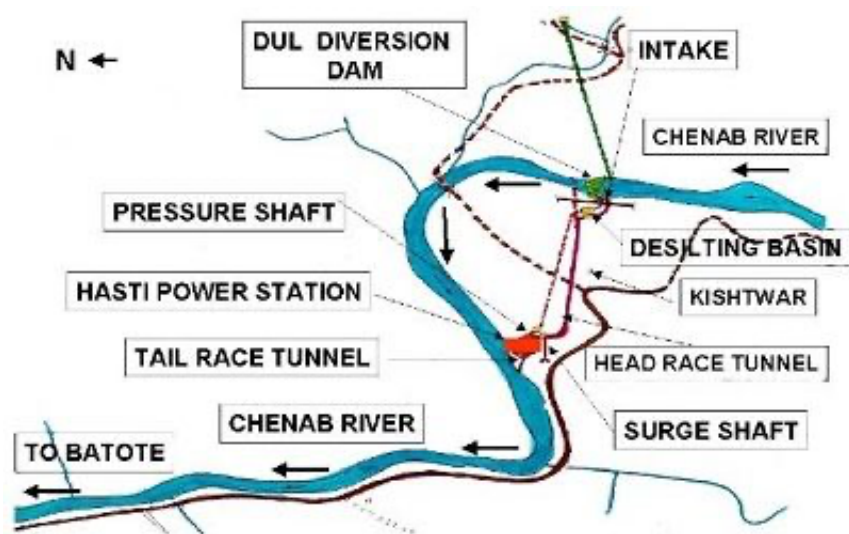
Location	Kishtwar (Jammu & Kashmir)
Approach	Nearest Rail Head: Udhampur Nearest Airport: Jammu / Srinagar
Installed Capacity	390 MW (3 × 130 MW)
Design Energy	1907 MU (90% dependable year with 95% machine availability)
Beneficiary States	Jammu & Kashmir, Punjab, Haryana, Chandigarh, Delhi, Rajasthan, Uttar Pradesh and Uttarakhand
Date of Commercial Operation (COD)	Unit-I : 07.04.2007 Unit-II : 07.04.2007 Unit-III : 07.04.2007
Dam Type	Concrete Gravity Dam
Dam Height	65m
Dam Length	186m
Full Reservoir Level (FRL)	EL. 1266.5 m
Minimum Draw Down Level (MDDL)	EL. 1238.9 m
Maximum Tail Water Level (TWL)	EL. 1032.593 m

3) SPECIFICATION (Continued)

Head Race Tunnel (HRT) Diameter	7.46 m
Head Race Tunnel (HRT) Length	10.571 km
Head Race Tunnel Type	Horse Shoe Shape
Penstock Diameter	3.46 m
Penstock Length	93 m (Circular Steel Lined)
Tail Race Tunnel (TRT) Diameter	7.46 m
Tail Race Tunnel (TRT) Length	298m
Tail Race Tunnel (TRT) Type	Horse Shoe Shape
Power House	Underground Power House having 3 generating units of 130 MW each
Turbine Type	Francis Turbine
Rated Head	207.5 m
Design Discharge	213 Cumecs (for all three units)
Number of Guide Vanes	24
Generator Rating	145 MVA
Rated Voltage	11 ± 10% kV
Number of Poles	20
Rated Speed	300 RPM Anti-
Direction of Rotation	clockwise(ViewedfromTop) 630
Runaway Speed	RPM

3) SPECIFICATION (Continued)

Generator Step-Up Transformer(GSU)	Single Phase Step-Up Transformer
Low Voltage (LV)	11kV
High Voltage (HV)	400 kV
Transformer Rating	48.33 MVA
Cooling Type	OFWF
GIS Rating	400 kV
Switchyard / Pothead Yard	Outdoor Pothead Yard
Power Evacuation	Through one 400 kV transmission line from Dulhasti to Kishenpur



5) DAM

The dam of the Dulhasti Hydroelectric Project is situated at a place called Dool in the Kishtwar district of Jammu and Kashmir. The type of dam used in the project is a concrete gravity dam, which has been designed to withstand enormous hydraulic pressure while ensuring structural stability and operational reliability. The dam serves as one of the most important components of the hydroelectric project and plays a vital role in water regulation and power generation.

The height of the dam above the deepest foundation level is 65 m and the length at the crest is 186 m. The top elevation of the dam is approximately 1270 m. The structure has been engineered to provide safe and efficient storage of water while maintaining long-term durability under varying operating conditions.

The spillway arrangement consists of four radial gates having an opening size of 9.0 m $\times$ 9.5 m. In addition, three ungated crest spillway bays measuring 11.6 m $\times$ 2.0 m are provided with a long chute for the safe discharge of excess flood water. These spillways help maintain reservoir levels and prevent overtopping during heavy inflow conditions.

There are four spillway gates available at the dam, each having dimensions of 9 m width and 9.5 m height with a radius of 13 m. These gates are operated by two 160-ton servomotors, ensuring smooth and reliable gate operation during normal as well as flood conditions.

The dam is also equipped with an upstream bulkhead gate consisting of five elements having a total weight of approximately 150 tonnes. The wheel-type bulkhead gate has dimensions of 9.0 m width and 13.0 m height and is primarily used during maintenance, inspection, and emergency shutdown operations.

Apart from facilitating hydroelectric power generation, the dam also contributes to effective water management and flood control. Its robust design, reliable operation, and strategic importance make it a key component of the Dulhasti Hydroelectric Project and a significant asset in harnessing the hydropower potential of the Chenab River.

6) RESERVOIR

The reservoir of the Dulhasti Hydroelectric Project plays an important role in storing water and ensuring a continuous supply for power generation. The Full Reservoir Level (FRL) of the reservoir is EL. 1266.5 m, while the Minimum Draw Down Level (MDDL) is EL. 1238.9 m. The maximum water level in the reservoir is EL. 1268.8 m.

The reservoir provides the necessary water storage required for efficient operation of the hydroelectric plant and helps regulate river flow during varying seasonal conditions. According to the reservoir elevation-area-capacity data, the gross storage capacity of the reservoir is approximately 11.55 million cubic meters, whereas the live storage capacity is about 10.97 million cubic meters.

The stored water ensures reliable power generation, supports effective water management, and contributes to the overall performance and stability of the Dulhasti Hydroelectric Project. The reservoir also assists in maintaining a steady water supply during periods of fluctuating river discharge, thereby enhancing the operational efficiency of the power station.

Thus, the reservoir serves as a vital component of the project by providing adequate storage capacity, improving operational flexibility, and ensuring uninterrupted hydroelectric power generation throughout the year. It also helps in maintaining reservoir levels during seasonal variations and supports the overall performance of the hydroelectric system.

7) INTAKE STRUCTURE

The intake structure has an invert level of EL. 1226.0 m. The project consists of two intake structures, one of which is provided for Dulhasti Stage-II. The intake opening is circular in shape with a diameter of 7.7 m, allowing the required quantity of water to enter the water conveyance system efficiently.

The intake gates are of size 5.5 m × 7.0 m and are operated by hydraulic cylinders. The lower gate is operated by a 124-ton hydraulic cylinder, the intermediate gate by an 81.3-ton hydraulic cylinder, and the upper gate by a 48-ton hydraulic cylinder. These gates regulate the flow of water entering the system and ensure safe operation under different operating conditions.

A stoplog gate consisting of three elements is also provided at the intake structure. The stoplog gate has dimensions of 5.5 m width and 7.0 m height and is operated using a 14-ton monorail hoist. The stoplog arrangement is mainly used during maintenance and inspection activities when isolation of the intake opening is required.

The intake structure serves as an important component of the Dulhasti Hydroelectric Project by controlling the entry of water into the head race tunnel and ensuring smooth and reliable operation of the power generation system. Its robust design contributes significantly to the safety, efficiency, and long-term performance of the hydroelectric project.

8) DESILTING CHAMBER

The type of desilting chamber used in the Dulhasti Hydroelectric Project is the Dufour type with a length of 240 m. The desilting chamber is designed to remove suspended silt particles from the water before it enters the head race tunnel, thereby protecting hydraulic equipment from excessive wear and erosion.

The project consists of two desilting chambers, each having dimensions of 15 m × 15 m. These chambers provide sufficient settling space for sediment particles and ensure cleaner water flow towards the power generation system.

Each desilting chamber is provided with a D-shaped silt flushing tunnel of size 4.5 m × 4.5 m. The flushing tunnel helps in the periodic removal of deposited silt and maintains the efficiency of the chamber during operation.

Two inlet gates are provided, each having dimensions of 4.8 m width and 4.8 m height. These gates regulate the flow of water entering the desilting chambers and ensure safe and controlled operation under different hydraulic conditions.

The desilting chamber plays a vital role in improving the operational life and performance of the hydroelectric project by reducing sediment-related damage to turbines and other hydraulic components.

9) GOVERNING SYSTEM

The governing system consists of a wicket gate and spherical valve control system, which operate independently and are controlled by an electronic governor. The governor is interlocked with the pumping unit, sump tank, air/oil pressure tank, and other necessary control units to ensure smooth and reliable operation of the generating unit.

The governing system operates at a pressure of 64 bar and plays an important role in regulating turbine speed, maintaining stability, and controlling power output under different operating conditions.

Its main functions are as follows:

1. Start-up and acceleration
2. Synchronization
3. Coupled unit operation
4. Return to speed no-load condition
5. Shut down

The governing system ensures efficient turbine control, safe operation, and stable performance of the generating unit during various stages of operation. It helps maintain the required speed and output of the turbine-generator set while ensuring reliable and secure plant operation.

10) HEAD RACE TUNNEL (HRT)

The Head Race Tunnel (HRT) is one of the most important components of the Dulhasti Hydroelectric Project. It conveys water from the intake structure to the underground powerhouse for power generation. The total length of the HRT is approximately 10.586 km and it has been designed to carry a discharge of 213 cumecs efficiently with minimum hydraulic losses.

The tunnel consists of two sections. The first section is of circular shape having a diameter of 7.7 m and a length of 2892 m. The second section is of horse-shoe shape with a diameter of 7.46 m and a length of 7694 m. The combination of these two sections ensures smooth water flow while maintaining structural stability throughout the tunnel alignment.

An air exhaust shaft is also provided in the HRT system for safe release of trapped air and pressure regulation. The shaft has a diameter of 500 mm (steel lined) and a depth of approximately 413 m. This arrangement helps in preventing air accumulation and ensures efficient tunnel operation under varying flow conditions.

The Head Race Tunnel plays a crucial role in delivering water to the turbines with the required head and discharge. Its robust design, large carrying capacity, and reliable operation contribute significantly to the efficient performance of the Dulhasti Hydroelectric Project.

11) SURGE SHAFT

The surge shaft used in the Dulhasti Hydroelectric Project is of the restricted orifice type. It is designed to protect the water conveyance system from sudden pressure fluctuations and water hammer effects caused by rapid changes in turbine load conditions.

The surge shaft has a diameter of 18.25 m and extends from the HRT invert level of EL 1171.5 m to a maximum surge level of EL 1286.0 m. The overall structure provides sufficient storage and pressure balancing capacity during transient operating conditions.

A fixed wheel type gate measuring 5.0 m × 9.0 m is installed at the surge shaft and is operated through a hydraulic hoist mechanism. The surge shaft plays an important role in maintaining hydraulic stability and ensuring the safe operation of the hydroelectric system.

12) PRESSURE SHAFT

The pressure shaft conveys water from the surge shaft to the underground powerhouse under high pressure. It is designed to withstand the hydraulic forces developed due to the high operating head of the project.

The project consists of one pressure shaft which is partly concrete lined and partly steel lined. The shaft has a diameter of 6.7 m and a vertical depth of approximately 138 m. The steel-lined section provides additional strength and reliability under high-pressure conditions.

The pressure shaft forms a vital link between the water conveyance system and the turbines. Its robust construction ensures efficient water delivery, minimizes hydraulic losses, and contributes to the reliable operation of the Dulhasti Hydroelectric Project.

13) POWER HOUSE

The powerhouse of the Dulhasti Hydroelectric Project is an underground structure located on the left bank of the Chenab River. It is designed to accommodate all major hydroelectric generating equipment and associated auxiliary systems required for efficient power generation. The underground arrangement provides structural safety, operational reliability, and protection from adverse environmental conditions.

The installed capacity of the powerhouse is 390 MW, comprising three generating units of 130 MW each. The powerhouse cavern has excavated dimensions of approximately 96 m × 19.8 m × 39.18 m, providing adequate space for installation, operation, and maintenance activities. A separate transformer cavern of dimensions 90.63 m × 15.5 m × 26.25 m is provided for housing transformers and associated electrical equipment.

The powerhouse is equipped with vertical Francis turbines, which are suitable for medium-head and medium-discharge hydroelectric applications. The centreline elevation of the machine is EL 1024.5 m, while the project operates under a maximum gross head of 237.0 m and a net head of 207.5 m.

The design discharge through the turbines is approximately 213 cumecs for all three generating units. The annual design energy generation of the project is about 1907 million units (MU) in a 90

14) TURBINE

In general, hydraulic turbines are classified according to the available head and discharge conditions. The commonly used turbine types are:

1. Francis Turbine – Medium head and medium discharge

2. Kaplan Turbine – Low head and high discharge

3. Pelton Turbine – High head and low discharge

The Dulhasti Hydroelectric Project utilizes a vertical shaft Francis turbine for power generation. Francis turbines are highly efficient and are well suited for medium-head hydroelectric projects. The turbine runner used in the project is of FN 52-13 blade configuration and is designed to operate efficiently under varying load conditions.

Each turbine unit has a rated discharge of approximately $69.4 \text{ m}^3/\text{s}$ and delivers a normal shaft output of about 132.4 MW. The turbine is equipped with 24 wicket gates that regulate the flow of water entering the runner. The maximum turbine efficiency is approximately 90

The Francis turbine guide bearing is self-cooled and self-lubricated. The turbine assembly incorporates combined thrust and guide bearing arrangements to ensure smooth operation, reduced vibration, and long service life. The turbine system plays a vital role in the efficient generation of electrical power at the Dulhasti Hydroelectric Project.

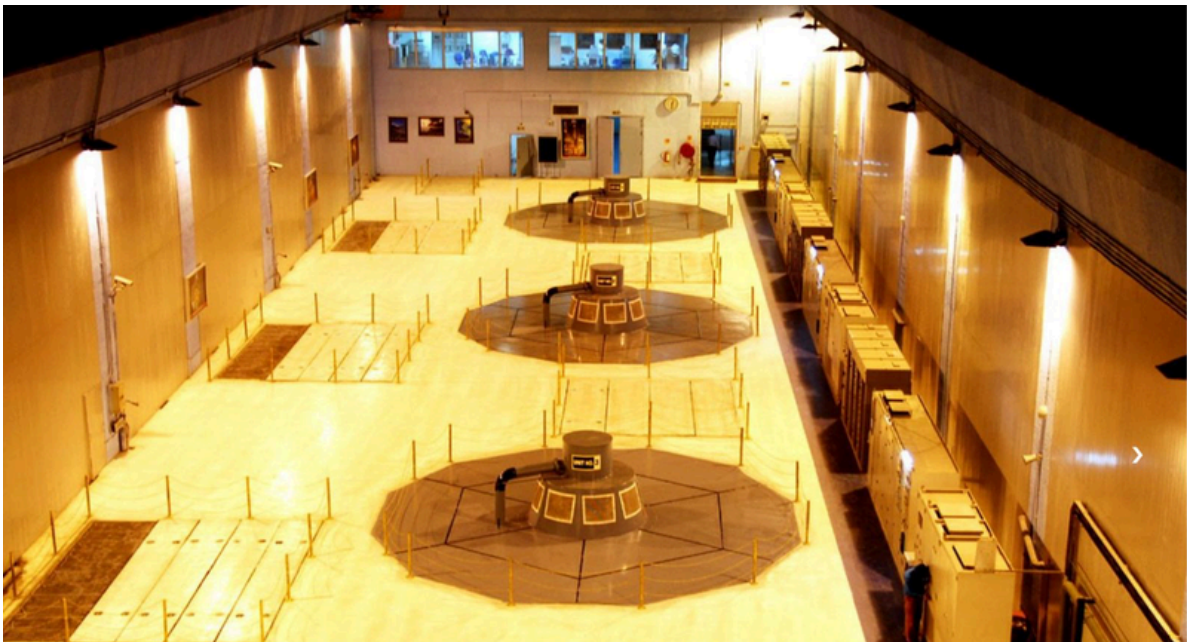


Figure 14.1: Turbine Hall View from Top



Figure 14.2: Francis Turbine

15) GENERATOR

Generator is a device that converts mechanical energy into electrical energy. Here it is used for generation of electricity by using stator as conductor and rotor as flux. When flux cuts the conductor it causes the flow of voltage into the conductor. In general generator is classified into three parts:-

- A) Stator
- B) Excitation
- C) Rotor

A) Stator:- It is the outer portion of the generator having a weight of 137 ton and weight of stator frame is 22 ton and it's core height is 1870 mm in the stator. The flow of current is 7611 A and 11 kV voltage. It is a type of concentric and double layer of windings.

B) Excitation:- Here the type of generator excitation is static and excitation voltage is 230 V. In the excitation, excitation current at nominal load is 1612 A which is a type of DC current.

C) Rotor:- It is the most important part of generator and it's moving phenomenon causes the generation of electricity. Rotor has a maximum rotation speed of 630 rpm but rated speed is 300 rpm.

16) DRAFT TUBE

The draft tube used in the Dulhasti Hydroelectric Project is a cone-shaped tube installed at the outlet of the turbine. Its primary function is to recover the kinetic energy of water leaving the turbine runner and convert it into pressure energy, thereby improving the overall efficiency of the turbine system.

The draft tube cone has a diameter of 2750 mm at the inlet and 3290 mm at the outlet. The total length of the cone section is 3001 mm. The distance between the vertical unit axis and the elbow outlet is approximately 4.3 m, ensuring smooth discharge of water into the tail race system.

The draft tube is designed to reduce the velocity of water exiting the turbine while maintaining continuous flow. This helps in minimizing energy losses and improving power generation efficiency.

The structure is equipped with one gate consisting of two elements. Each element has dimensions of 5.28 m $\times$ 2.425 m $\times$ 0.8 m. These gate elements are used for maintenance and inspection purposes and provide safe isolation of the draft tube whenever required.

Due to its robust design and efficient energy recovery characteristics, the draft tube plays an important role in enhancing the performance and operational reliability of the hydroelectric generating unit.

17) COMPUTER & CONTROL SYSTEM

The computer and control system used in the Dulhasti Hydroelectric Project provides centralized monitoring and control of all machines and equipment from the main control room. The entire system operates through a package known as Centralog, which enables efficient supervision of plant operations and ensures reliable communication between different units.

The language used for controlling is “C” and the system consists of 2 data servers, 6 control units, and 2 operator consoles with 3 log units. These components work together to collect, process, store, and display operational data from various sections of the power plant.

The data bus used in the system is Contronet Redundant Data Highway, which provides secure and uninterrupted communication between all control devices. The redundancy feature improves reliability and minimizes the possibility of communication failure during plant operation.

This computer and control system plays a vital role in ensuring safe operation, quick fault detection, real-time monitoring, and efficient power generation at the hydroelectric station.

18) MECHANICAL WORKSHOP

The Mechanical Workshop plays an important role in maintaining and improving the performance, reliability, and service life of various machines and equipment used in the hydropower project. Regular inspection, repair, and servicing activities are carried out in the workshop to ensure smooth plant operation. The major machines and equipment available in the workshop are described below.

A) JCB

The term JCB stands for Joseph Cyril Bamford. It is a hydraulically operated multipurpose machine widely used for excavation, loading, material handling, and maintenance activities. The machine available in the workshop has an engine power of approximately 180 HP and is equipped with a four-cylinder engine. Engine oil is used as the primary lubricant. Different sizes of JCB machines can be employed depending upon the nature and scale of work to be performed.

B) Caterpillar Machine

Caterpillar equipment is mainly used for surface leveling, road cutting, earthmoving, and heavy-duty construction work. The machine is manufactured by BEML and has an engine power of about 380 HP. It is equipped with a six-cylinder engine and a turbocharger for improved performance. The machine weighs approximately 44 tonnes and uses a 25-plate, 12-volt battery system. A hydraulic tank containing Servo-68 oil is used for generating hydraulic pressure required for operation.

C) Snow Cutter

A Snow Cutter is a specialized machine used for removing accumulated snow from roads and operational areas during winter conditions. The machine collects and throws snow to the roadside, thereby maintaining accessibility and ensuring safe transportation. It is equipped with a four-cylinder engine and incorporates a jack and servo motor system for effective control and operation. A rotary pump is also installed in the machine for efficient functioning.

D) Grinder Machine

A Grinder Machine is a power-driven machine tool used for grinding and finishing operations. It employs an abrasive wheel as the cutting tool and is widely used for sharpening tools, removing excess material, improving surface finish, and achieving precise dimensions. During grinding, each abrasive grain on the wheel acts as a small cutting edge that removes tiny chips from the workpiece through shear deformation. Grinding operations are essential for maintaining dimensional accuracy and surface quality of engineering components.

The Mechanical Workshop therefore serves as a vital support facility for the hydropower plant by ensuring the availability, maintenance, and repair of critical machinery required for construction, operation, and maintenance activities throughout the project area.

19) CONCLUSION

The internship at Dulhasti Power Station was a valuable learning experience that helped me understand the practical working of a hydroelectric power plant. During the training period, I gained knowledge about the major components of the project such as the dam, reservoir, intake structure, desilting chamber, head race tunnel, surge shaft, pressure shaft, powerhouse, turbine, generator, draft tube, computer and control system, and mechanical workshop.

The visit provided an opportunity to observe how electrical and mechanical systems work together for efficient power generation. I also learned about the operation, maintenance, safety practices, and monitoring systems used in a modern hydroelectric project.

This training enhanced my technical knowledge and gave me practical exposure to equipment and processes that are difficult to understand only through classroom studies. It also improved my understanding of renewable energy generation and the importance of hydropower in meeting the growing energy demands of the country.

Overall, the internship was highly informative and beneficial. It strengthened my interest in mechanical engineering and provided valuable industrial experience that will be helpful in my future academic and professional career.